

Supply chain issues in SME food sector: a systematic review

Supply chain
issues in SME
food sector

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Abstract

Purpose – The purpose of this paper is to investigate the significant supply chain issues in the small and medium-sized enterprise (SME) food industry. The objectives are to identify the major themes and the dynamic evolution of SME food supply chain (FSC) issues, the current research trends, the different modelling approaches used in SME FSC, and the most addressed SME food sector.

Design/methodology/approach – In all, 3,733 published articles from 2002 to 2018 in the Clarivate Analytics Web of Science database were collected, from which 1,091 articles were shortlisted for the review. The authors used bibliographic coupling combined with co-word analysis to identify the historical relations of the research themes that emerged during the periods 2002–2014 and 2002–2018.

Findings – This research identified five major research themes such as production and distribution in alternative food networks, relationship, safety and standards in the FSC, greenhouse gas (GHG) emission impact of the farm food system, traceability and product quality in FSC and asymmetric price transmission in the FSC. Among the identified themes, GHG emission impact of the farm food system and traceability and product quality in the FSC have received increasing attention in recent years. The dairy sector is the most addressed sector (36 per cent), followed by fruits and vegetables (27 per cent), meat and poultry (18 per cent), seafood (10 per cent) and grains and oilseed (8 per cent). It is also identified that the dairy sector has received significant attention in the “GHG Emission impact of farm food system” theme. Similarly, meat and poultry sectors have received much attention in the “Traceability and product quality in the food supply chain” theme. Also, the authors identified that the empirical modelling approaches are the most commonly used solution methodology, followed by the conceptual/qualitative methods in the SME FSC.

Originality/value – This study maps and summarizes the existing knowledge base of supply chain issues in the SME food sector. The results of this review provide the major research areas, most commonly used approaches and food sectors addressed. This study also highlights the research gaps and potential future research direction.

Keywords SME, Literature review, Supply chain, Dairy, Meat, Fruit and vegetables

Paper type Literature review

1. Introduction

The food supply chain (FSC) technically spans the entire “farm-to-fork” process. A typical FSC includes farmer, marketer who supplies products to food processors/producers, wholesaler, retailers/caterer and consumer (Mattevi and Jones, 2016a). However, the exact supply chain path for a particular food product depends on the product characteristics, size and market power of the supply chain members (Dani and Deep, 2010). Managing FSC is complex as compared to other supply chains due to the factors like the freshness of food, quality, safety, etc. (Zhong *et al.*, 2017). A major portion of the food industries participating in the FSC comprises of small and medium-sized enterprises (SMEs) (Mertins *et al.*, 2012). The annual statistics of Food Drink Europe (2016) shows that about 99.1 per cent of all food processing industries in the EU are SMEs. The food processing SMEs comprise dairy, beverages, food chains, spice trading, meat, fruits, vegetables, etc. In most of the developing countries, including India, food SMEs provide not only huge employment opportunities but they also ensure regional balance by taking industrialization to rural and backward areas. The evaluation and advancement of FSCs from SMEs are essential, from the point of view of SME performance and for the health of the society (Henchion and McIntyre, 2005). With the increasing digital technologies, designing/restructuring

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the FSC involving SMEs is a challenging task. The SMEs of the FSC networks should work collaboratively to deliver high-quality food to consumers (Mertins *et al.*, 2012). Prakash (2018) investigated the food processing supply chain (FPSC) in the bilateral context of the UK and India and reported that the UK FPSC is highly integrated with a high degree of consolidation in upstream and downstream stages, whereas the Indian FPSC is long and fragmented.

In the last few years, many food quality and safety issues are frequently reported. For instance, the milk scam disclosed in India found that saturated fatty oil and powder were blended with the milk to increase the sale (Dandage *et al.*, 2017). The FSC has received increasing attention from both academia and practitioners, as consumers are concerned about food quality and safety (Mahajan *et al.*, 2017). For a better understanding of this research domain, it is important to map and summarize the existing knowledge base through an effective literature review. However, reviews on this research area were focussed only on a specific food sector (Ganeshkumar *et al.*, 2017; Beg *et al.*, 2017; Denham *et al.*, 2015) or a particular issue like food safety (Auler *et al.*, 2017; Manning and Soon, 2016), food traceability (Dandage *et al.*, 2017), environmental issues (Borodin *et al.*, 2016; Denham *et al.*, 2015; Wiedemann *et al.*, 2015a), perishable food products (Siddh *et al.*, 2015) and processed FSC (Prakash, 2018; Mahajan *et al.*, 2017). It is observed that none of the previous literature map and summarize the existing knowledge base in the SME FSC domain (Luo *et al.*, 2018). Most of the past reviews on the FSC are narrative as shown in Table I. A systematic literature review (SLR) is preferred, as it outperforms other review methods since its characteristics, explanatory, heuristic nature, transparency and inclusivity, eliminate any errors and bias occurred, thereby making the research result more objective (Denyer and Tranfield, 2009). To address this research gap, we present a SLR of SME FSC. A total of 1,091 research articles published from 2002 to 2018 are collected from the Web of Science (WoS) database. We use bibliographic coupling (BC) combined with co-word analysis to identify the historical relations of the research themes that emerged during the periods 2002–2014 and 2002–2018. The broader thematic context is obtained for the period 2002–2014, whereas the inclusion of recent articles (i.e. 2015–2018) resulted in identifying the emerging research directions out of these broader issues.

Sl No.	Author(s)	Research focus			Review type	
		Supply chain	SMEs	Food sector	NLR	SLR
1	Prakash (2018)	*		*		*
2	Mahajan <i>et al.</i> (2017)	*		*	*	
3	Beg <i>et al.</i> (2017)	*	*	*	*	
4	Gharehgozli <i>et al.</i> (2017)	*		*	*	
5	Notarnicola <i>et al.</i> (2017)	*	*	*	*	
6	Dandage <i>et al.</i> (2017)	*	*	*	*	
7	Zhong <i>et al.</i> (2017)	*		*	*	
8	Fassam and Dani (2017)			*		*
9	Ganeshkumar <i>et al.</i> (2017)	*	*	*	*	
10	Auler <i>et al.</i> (2017)	*		*		*
11	Mahajan <i>et al.</i> (2017)	*		*	*	
12	Manning and Soon (2016)	*		*	*	
13	Fraser <i>et al.</i> (2016)	*		*		*
14	Borodin <i>et al.</i> (2016)	*	*	*	*	
15	Sjauw <i>et al.</i> (2016)	*	*	*	*	
16	Berti <i>et al.</i> (2016)	*	*	*	*	
17	Siddh <i>et al.</i> (2015)	*		*		*
18	Wiedemann <i>et al.</i> (2015a)	*	*	*	*	
19	Denham <i>et al.</i> (2015)	*		*	*	
20	Fredriksson and Liljestrand (2015)	*		*	*	

Table I.
Recent reviews on
food supply chain

Notes: NLR, narrative literature review; SLR, systematic literature review. *Highlights the research focus and review type reported

This paper is organized as follows. Section 2 presents the proposed review methodology and the steps followed for locating the study. Result and discussion of BC are elaborated in Section 3. Finally, Section 4 concludes the paper by highlighting current research focus and future research directions.

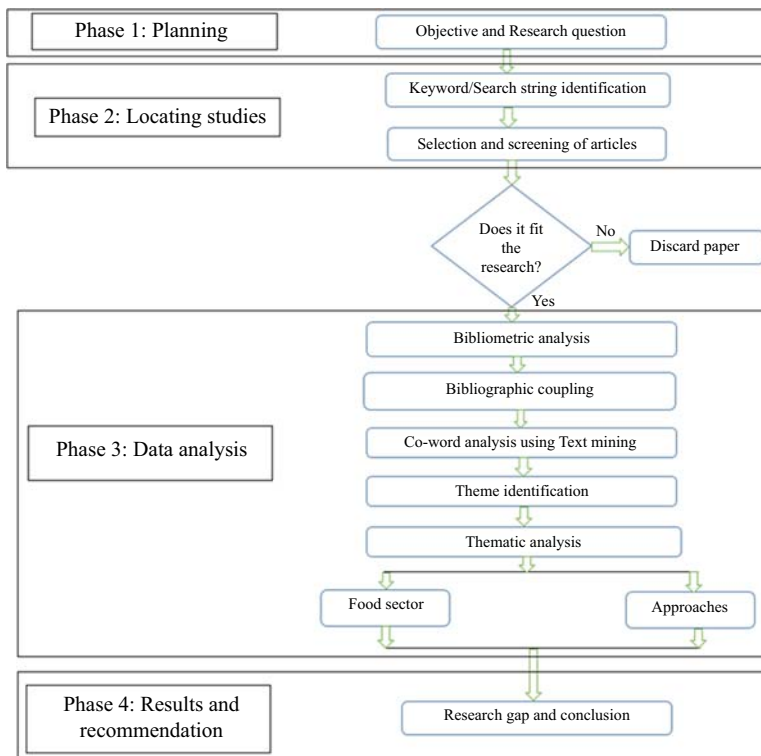
2. Review methodology

A SLR methodology is proposed to identify the major supply chain issues in SME food industry. A four-phase SLR methodology (see Figure 1) is applied following the work of Denyer and Tranfield (2009).

2.1 Research design

The methods of BC, co-word and network analysis are used to answer the following four research questions, which are appropriate and influence the goal and focus of the review:

- RQ1. What are the major supply chain issues/themes addressed in the SME food industry?
- RQ2. What are the current research trends in the SME FSC?
- RQ3. What are the different modelling approaches used in SME FSC?
- RQ4. Which food sector has received attention in the literature?



Source: Adapted from Denyer and Tranfield (2009)

Figure 1.
Review method

2.2 Locating studies and evaluation

The articles for the study are identified and screened after formulating the research questions. This section summarizes the steps followed for locating the study and its evaluation.

2.2.1 Time horizon. Articles published during 2002–2018 were considered for the study. The works prior to 2002 were excluded due to the limited number of publications on the supply chain issues in SME food sectors (Mahajan *et al.*, 2017; Siddh *et al.*, 2015). Being the immediate previous year at the time of the review, we included all articles till the end of the year 2018.

2.2.2 Selection of database. WoS database from Clarivate Analytics is used to collect the articles. We selected the WoS database for the following solid reasons. The first reason is to get a comprehensive result and to ensure the quality of articles. The second WoS database includes more than 18,000 journals of high impact from six different citation databases. Third, the retrieved result from WoS can be organized and analysed easily through bibliometric software tools (Colicchia and Strozzi, 2012).

2.2.3 Search criteria for article selection. As proposed by Fahimnia *et al.* (2015), a three-level keyword was used to extract articles as shown in Table II. “OR” connectors were used to connect keywords in each level, and different levels of keywords were connected by using “AND” connectors. To find root word plus, we used “*” symbol at the end of some words. Different terms with their variants were used at each level of the search strings so as to not limit the database search to a specific aspect of the research domain, as our primary objective was to identify the major research concepts. For example, Level 1 keywords used the term “supply chain” including its variants such as “inventory”, “logistics”, “supply network” or “value chain”.

2.2.4 Sorting of articles. In order to include the most relevant articles, a set of inclusion and exclusion criteria is defined, followed by Newbert (2007) as listed below:

- (1) peer-reviewed scientific journals published only in English are included;
- (2) articles published during 2002–2018 were considered for the study since articles that met our selection criteria before 2002 were very limited;
- (3) selected articles that contain at least one keyword in each level of the search string in the title, abstract and keywords;
- (4) commercial publications, book chapters, proceedings papers, periodicals and working papers are not included; and
- (5) articles that are not relevant to the SME FSC are excluded.

During the initial search, 3,733 articles are retrieved from the WoS database, and after applying the inclusion and exclusion criteria (1–4), a total of 3,040 articles are retained. Both the authors independently screened the papers on the basis of the title, abstract and keywords. The papers for which both the authors agreed have been included. In case of any discrepancy, we included a paper for full-text reading. Specifically, we excluded 1,949

Table II.
Keyword string

Level 1	“supply chain*” or “inventory” or “logistic*” or “supply network” or “value chain”
Level 2	“small business*” or “sme” or “SMEs” or “small and medium” or “msme” or “small firm*” or “small industr*” or “small scale industr*” or “micro business” or “household business” or “small organisation*” or “small compan*” or “small entit*” or “medium compan*” or “small enterpri*” or “medium enterpri*” or “family business” or “small retail*” or “small to medium” or “farm*”
Level 3	“*food” or “beverage*” or “dairy” or “fruit*” or “vegetable*” or “meat” or “beef” or “fish” or “halal” or “drinks”

articles as irrelevant (criteria 5) with a focus on food and bioenergy (biofuel), the production process of food, animal welfare, etc., and the remaining 1,091 articles were then shortlisted for further appraisal.

We adopted a comparative approach to the analysis of the data, in which we divided all papers into two roughly equal parts (Vogel and Güttel, 2013). Initially, we performed an analysis of 377 research papers published between 2002 and 2014. Then, we added the rest of 714 articles and repeated the same analysis from the period 2002 to 2018. By comparing the result of these two-time sliced articles, it is possible to find the research direction and the contribution of the recent research articles to the literature on SC in SME food industry.

2.3 Bibliographic coupling (BC) network and cluster analysis

Bibliometric and network analysis tools are used in a systematic review to identify and cluster significant research concepts. A bibliometric network is a weighted network consisting of nodes and edges. The node represents a knowledge actor like a research paper, authors, journal, authors' institute/country or keywords. The edge between two nodes denotes the relationship between them. The five commonly studied edge relationship includes citation, co-citation, BC, co-author and co-word analysis (Zupic and Cater, 2015). A BC connects journals, authors or documents with respect to the number of shared references. Among the bibliometric networks, BC could capture the research dynamics from past to present, as it includes the most recent contributions (Vogel and Güttel, 2013). The bibliographic strength between the two articles is defined by the number of items shared by these two articles in their reference records (Zhao and Strotmann, 2008). Boyack and Klavans (2010) compared the accuracy of co-citation analysis, BC, direct citation and found that BC outperforms the other two. In literature, bibliographic analysis tools like CiteSpace, Pajek, Bibexcel, R, SciMat and Saint are proposed for bibliometric analysis (Cobo and Herrera, 2011). Among these, BibExcel allows processing the different types of file formats like Medline format, Scopus RIS format and the WoS plain text format (Persson *et al.*, 2009). Also, Bibexcel allows data modification or filtering of imported details from various databases such as WoS and Scopus (Bjurström, 2011). Due to these advantages, BibExcel is used in this research for the bibliometric analysis.

In order to conduct BC, the plain text format from WoS containing the details of author, the title of publication, year of publication, abstract and authors affiliation from the shortlisted 1,091 articles are exported to Bibexcel software. To further map and visualize, the BC among the 1,091 articles network and cluster analysis tools are used. In literature, UCINET, Mathematica, Pajek, VOS Viewer and Gephi are the commonly used network analysis tools. Among these, Gephi is generally used in research projects, given its simplicity (Fahimnia *et al.*, 2015). Gephi is capable of handling large network data and helps in filtering, navigating, manipulating and clustering the data. It can also handle a variety of graph data sets generated in comma-separated values files, Graph and Net file formats. In this research, the shared references from Bibexcel in .net format were imported to Gephi and were used for networking and clustering the research articles.

To identify the sub-fields/themes of a research domain, the nodes of a network are divided into clusters or modules using clustering techniques (Fahimnia *et al.*, 2015). The degree of association is strong among members of the same cluster and weak between members of different clusters. For dividing the network into clusters, the default Louvain clustering algorithm in the Gephi software is used, which will maximize the modularity index by iterating each node to all clusters (Rajagopal *et al.*, 2017). Modularity is an important metric that provides information about the overall clustering of the networks. The modularity index is a scalar value between -1 and $+1$ that measures the density of links inside the clusters vs the links between the clusters (Fahimnia *et al.*, 2015).

2.4 Text mining (TM)

To find out the research focus of each sub-field, the contents of either all or the lead papers in each sub-field are analysed and evaluated. The interpretation and analysis of sub-field depend greatly on subjective factors and require expertise in the specific research field. To avoid the reviewer bias and for a deeper understanding of the concepts addressed in each sub-field, researchers have further used text mining (TM) as a content analysis technique to retrieve the co-occurred words from the documents and to build a conceptual structure of the domain (Ghadge *et al.*, 2012; Song and Chambers, 2014). In this research, the co-occurred words from the research articles in each sub-field are retrieved using TM to identify research concept precisely. Co-word analysis presumes that a group of aggregated terms/ words could indicate the underlying themes and that co-occurrences of terms/words could show the associations with the underlying themes. Title and abstract of all articles in each sub-field are downloaded from the WoS database in .txt format and text mined separately using Stanford NLP core 3.6.0. StanfordcoreNLP 3.6.0 software is used in this research for TM since it is flexible and simple (Song and Chambers, 2014).

Using Stanford metrics conversion Handler Programme, the output extensible markup language file from the Stanford NLP is converted to a GraphML network file from the identified co-occurred words. The GraphML file is then used as input in Gephi for visualization and further analysis of the co-word network. After establishing a co-word network, the research concept of each cluster is then identified by measuring the relative frequency of these co-occurred words and by measuring the network properties based on node and network-level metrics (Barbara and Wichmann, 2016). The node-level metrics can be divided into local: degree centrality, and the ones relative to the rest of the network (i.e. based on node's location within a network): closeness, betweenness and eigenvector centrality (Milojevic, 2013). The number of direct links between a node and other nodes in the cluster is defined by degree centrality. A word with a higher degree will be in the centre of the network and considered as the information hub. The closeness centrality is defined by the inverse of the average length of the shortest paths to or from all the other nodes in the network. Higher closeness centrality refers to higher influence on other nodes. The betweenness centrality is a measure of how often a node is located on the shortest path between other nodes in the network. Eigenvector centrality is based on the principle that importance of a node in a network may increase with having links to the other nodes that are themselves important. It is calculated by providing a relative score for each node proportional to the sum of the scores of its neighbours (Bonacich, 2007). The equations used to calculate the node centrality are given in Appendix 1.

3. Results

3.1 Bibliometric analysis

The time period distribution of 1,091 articles is shown in Figure 2. It is noted that the number of articles was limited to 377 during the period 2002–2014, and the remaining

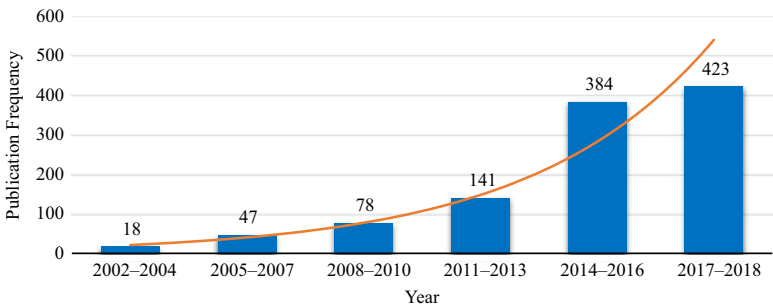


Figure 2.
Distribution of
reviewed papers by
year (up to
December 2018)

714 articles were published in the last four years, showing a sharp rise in the number of research publications. From the year 2002, there were a moderate number of articles; however, the time period 2011–2018 witnessed a significant growth in the number of articles. The spike in the research articles from the year 2011 can be attributed to the transdisciplinary and cross-disciplinary journals such as *Sustainability*, *Agricultural Systems*, *Journal of Cleaner Production* and *International Food and Agribusiness Management Review* that encouraged the publication in the FSC domain.

The number of reviewed articles published in leading journals is shown in Table III. Among the leading journals, *British Food Journal*, *Supply Chain Management – An International Journal* and *Food Policy* have published more number of papers during the period 2002–2014. However, during the period 2015–2018, the *Journal of Cleaner Production*, *British Food Journal*, *Sustainability*, *Agricultural Systems*, *Food Control* and *International Food and Agribusiness Management Review* have contributed significantly. It is observed that the *Journal of Rural Studies* is among the top contributing journals with publications related to alternative and local food network, even though it is of limited relevance to the research domain.

The distribution of reviewed papers by countries is shown in Table IV. It is found that most of the research articles are from developed countries where the USA accounts for 19.25 per cent of the publications, making it the country with the most publications globally, followed by the UK (10.82 per cent), Italy (10.45 per cent), Germany (7.15 per cent), Australia (6.87 per cent) and the Netherlands (6.14 per cent). Among the developing countries, China (5.41 per cent) and India (4.86 per cent) lead on the publications. Developed countries have the advanced infrastructure, availability of funds and research environment, whereas the limited funds, infrastructure and awareness result in a very few research publications in developing countries (Siddh *et al.*, 2015).

The frequently used keywords are shown in Table V. Since authors use different words to describe the same meaning, keywords often should be standardized. By following Su and Lee (2010), the keywords are standardized in such a way that plural words are converted to singular form, and similar words such as “SCM”, “value chain”, “supply chain management” are standardized to “supply chain management” on the basis of the popularity.

3.2 Bibliographic coupled network for the period 2002–2014

The node in a bibliographic coupled network refers to the citing documents, and the edges represent the number of references shared by any two nodes. Following the work of Vogel and Güttel (2013), we included only those articles that had at least four couplings with a minimum of two other articles to obtain more clarity. Through this procedure, the number of articles is reduced from 377 to 66 most interconnected papers. The input files in .csv format

Journals	2002–2014	2015–2018	Total
<i>Journal of Cleaner Production</i>	6	58	64
<i>British Food Journal</i>	15	42	57
<i>Sustainability</i>	2	31	33
<i>Food Control</i>	8	19	27
<i>Food Policy</i>	9	15	24
<i>Agricultural Systems</i>	4	20	24
<i>International Food and Agribusiness Management Review</i>	5	18	23
<i>Journal of Rural Studies</i>	9	9	18
<i>International Journal of Life Cycle Assessment</i>	6	12	18
<i>Supply Chain Management – An International Journal</i>	11	4	15

Table III.
Distribution of
reviewed papers
by journal

Table IV.
Global contribution
of authors

Countries/regions	No. of articles	Percentage
USA	210	19.25
UK	118	10.82
Italy	114	10.45
Germany	78	7.15
Australia	75	6.87
The Netherlands	67	6.14
Peoples R China	59	5.41
Canada	55	5.04
India	53	4.86
France	52	4.77
Belgium	38	3.48
Spain	35	3.21
Brazil	33	3.02
Scotland	30	2.75
South America	29	2.66
Kenya	29	2.66
Sweden	27	2.47
Indonesia	17	1.56
Portugal	13	1.19
Pakistan	8	0.73

Table V.
Top 10 keywords

Keyword	Frequency
Supply Chain Management	180
Life Cycle Assessment	89
Food Quality	84
Agriculture	77
Sustainability	51
Food Waste	41
Food Safety	39
Food Industry	31
Traceability	23
Carbon Footprint	23

are imported into Gephi, and a raw network of 66 nodes and 599 edges is created with a random positioning of the nodes. By applying the default Louvain algorithm in the Gephi software, the bibliographic coupled articles are grouped into four clusters, as shown in Figure 3. The modularity index of the network is 0.398, indicating a strong relationship between the nodes within and between the clusters (Rajagopal *et al.*, 2017).

Clusters 1 and 2 have received much attention, and they include 25 and 24 papers, respectively, among the 66 articles. Clusters 3 and 4 have received limited attention, as they include only 10 and 7 papers, respectively. The number within each node denotes the article ID (see Appendix 2). The size of nodes in each cluster is proportional to their degree centrality. The top five articles (with article ID in curly bracket) in each cluster, based on at least one of the four measures of centrality, are given in Table VI. The influence of an article in a cluster can be identified from its degree, closeness, betweenness and eigenvector centrality. The article with a higher degree centrality exerts a heavier influence on the cluster and acts as an information hub. For example, the research work of {89} Freidberg *et al.* (2011) and {17} Ilbery *et al.* (2006a) are identified as the lead articles of Cluster 2 with degree centrality of 35 and 34, respectively, which indicates that these articles share

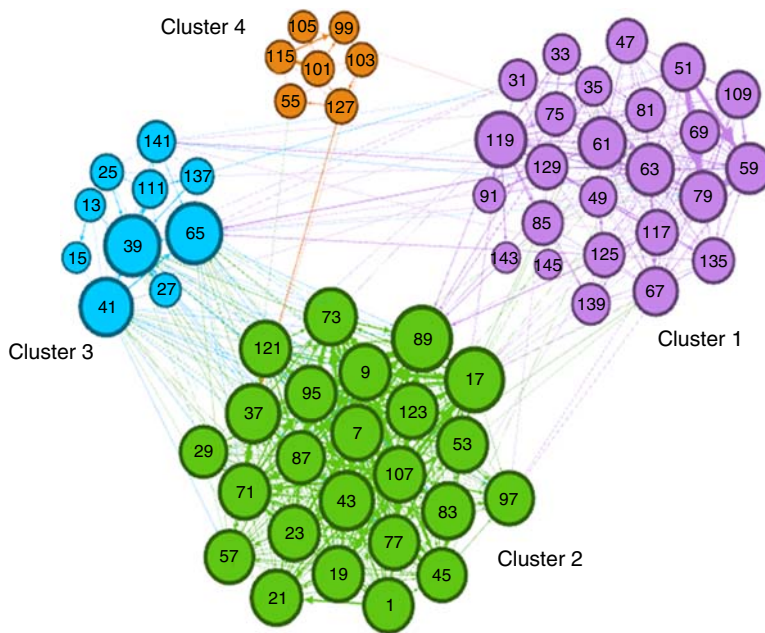


Figure 3.
Clustering of the
bibliographic coupled
network (2002–2014)

Cluster	Article[ID]	Degree	Closeness	Centrality	
				Betweenness	Eigenvector
1	Reardon <i>et al.</i> (2012) [119]	28 [1]	0.5666 [1]	210.72 [1]	0.3863 [1]
	Blandon <i>et al.</i> (2010) [79]	21 [4]	0.5312 [4]	25.4390	0.3083 [2]
	Minten <i>et al.</i> (2009) [61]	23 [2]	0.5440 [2]	47.14 [5]	0.2977 [3]
	Dries <i>et al.</i> (2009) [63]	23 [3]	0.5440 [3]	49.95 [3]	0.2945 [4]
	Amekawa (2009) [67]	19	0.5037	25.1237	0.2852 [5]
	Neilson (2007) [29]	21 [5]	0.5312 [5]	24.5658	0.283423
	Dries <i>et al.</i> (2014) [129]	17	0.527132	97.33 [2]	0.255
	Gorton <i>et al.</i> (2007a) [35]	13	0.503704	47.35 [4]	0.177016
2	Freidberg <i>et al.</i> (2011) [89]	35 [1]	0.6415 [1]	216.88 [2]	1 [1]
	Ilbery <i>et al.</i> (2006a) [17]	34 [2]	0.6238 [2]	162.59 [3]	0.9990 [2]
	Higgins <i>et al.</i> (2008) [43]	29 [3]	0.5762 [5]	39.92 [4]	0.9379 [3]
	Peter <i>et al.</i> (2010) [73]	27 [5]	0.561983	11.92	0.9136 [4]
	Risku Norja <i>et al.</i> (2008) [37]	29 [4]	0.5913 [3]	333.50 [1]	0.9133 [5]
	Miewald <i>et al.</i> (2013) [121]	26	0.5862 [4]	20.18	0.895454
	Connell <i>et al.</i> (2008) [45]	24	0.548387	38.54 [5]	0.80648
	Croom <i>et al.</i> (2007) [27]	27 [4]	0.5913 [3]	30.25 [5]	0.9057 [1]
3	Kottila <i>et al.</i> (2008) [39]	31 [1]	0.6071 [1]	125.37 [3]	0.8916 [2]
	Fischer <i>et al.</i> (2009) [65]	31 [2]	0.6071 [2]	136.75 [2]	0.8512 [3]
	Leat <i>et al.</i> (2008) [41]	28 [3]	0.5862 [4]	192.25 [1]	0.7931 [4]
	Zander <i>et al.</i> (2014) [141]	12 [5]	0.4892 [5]	35.60 [4]	0.1918 [5]
	Mueller <i>et al.</i> (2014) [127]	6 [1]	0.3976 [1]	59.49 [2]	0.0442 [1]
4	Van der Werf <i>et al.</i> (2009) [55]	5 [3]	0.3908 [2]	40.34 [4]	0.0431 [2]
	Flysjo <i>et al.</i> (2011) [103]	5 [4]	0.3906 [3]	68.28 [1]	0.0430 [3]
	Tarawali <i>et al.</i> (2011) [105]	3	0.3617 [4]	46.64 [3]	0.0127 [4]
	Christie <i>et al.</i> (2011) [101]	6 [2]	0.3177 [5]	14.60 [5]	0.0102 [5]
	Christie <i>et al.</i> (2012) [115]	4 [5]	0.290598	0.66	0.006975

Table VI.
The values and ranks
[in bracket] of top 5
articles based on at
least one of the four
measures of centrality
during the period
2002–2014

common references with papers of this cluster and also with a number of other articles in the whole network.

Closeness centrality is associated with a node's autonomy or independence in the whole network. A node with higher closeness centrality indicates that it is less reliant on other nodes and has a higher capacity for independent actions. From the results, we found that the works of {89} Freidberg *et al.* (2011), {17} Ilbery *et al.* (2006a), {39} Kottila *et al.* (2008) and {65} Fischer *et al.* (2009) have a high closeness centrality value, indicating the extensivity of their influence in the whole network. A node with high betweenness centrality specifies that it links various other nodes in the research domain, indicating that they play a more important role in the whole network than in the focal cluster. For example, the works of {37} Risku Norja *et al.* (2008), {89} Freidberg *et al.* (2011), {119} Reardon *et al.* (2012) and {41} Leat *et al.* (2008) have a high betweenness centrality value and serve as a bridge in the whole network connecting different clusters. From Table VI, it is clear that the articles from Clusters 2 and 3 have a higher betweenness centrality as compared to Cluster 4, implying that the research concepts of Cluster 4 have less association with other clusters.

Eigenvector centrality considers whether a node has many neighbours or important neighbours or both. From Table VI, higher eigenvector centrality is observed for the articles in Cluster 2, followed by {27} Croom *et al.* (2007), {39} Kottila *et al.* (2008) and {65} Fischer *et al.* (2009) of Cluster 3. Cluster 4 has the least eigenvector centrality in which the lead articles have an eigenvector centrality of 0.0442, indicating that the research issue of these clusters has less influence in the network. To identify the research theme in each cluster, we further conducted a TM technique and established a co-word network as shown in Figure 4. After establishing the co-word network, the research concept of each cluster is then identified by measuring the network properties based on degree, closeness, betweenness and eigenvector centralities (Milojevic, 2013).

The top 10 words in each cluster are identified by measuring the relative frequency of co-occurrence and centralities as presented in Table VII. The degree measure reported in Table VII shows that market economics of contract farming in SME FSC, alternative food network (AFN), supply chain relationship in the food industry and land use and emission impact of livestock systems are the research concepts of Clusters 1, 2, 3 and 4, respectively. It is observed from Table VII that the keywords Supply, Chain, Farmer, Food, Market and System are found in more than one cluster. These are bridging keywords that highlight the association between the clusters.

It is also observed that Clusters 1 and 2 have more bridge keywords with high betweenness centrality, indicating their close association with other clusters. In Cluster 4, only one bridge keyword is identified from among the top 10 keywords, indicating that this sub-field has less association with the other clusters. It is worth noting that this bridge keyword has a high eigenvector centrality in few clusters and low eigenvector centrality in other clusters, indicating its importance within the clusters. The keyword "SME" appeared in Clusters 1 and 2 within the top 10, indicating its strong association with both the clusters, whereas it appeared at the 17th position with low centralities in Cluster 3, indicating its weak association with the theme of Cluster 3. It is also observed that the word "SME" did not appear in Cluster 4 during this period.

3.2.1 Bibliographic network during 2002–2018. To trace the dynamic evolution of the research domain, we repeated the analysis after adding the research articles published from 2015 to 2018. Since the data are almost doubled, we limited to at least seven couplings with minimum of two other articles resulting in the number of included articles to 228 (see Appendix 3). By applying the default Louvain algorithm in Gephi, the 228 bibliographic coupled articles are grouped into five clusters as shown in Figure 5. The modularity index of the network is 0.429, indicating a strong association between the nodes within and between the clusters.

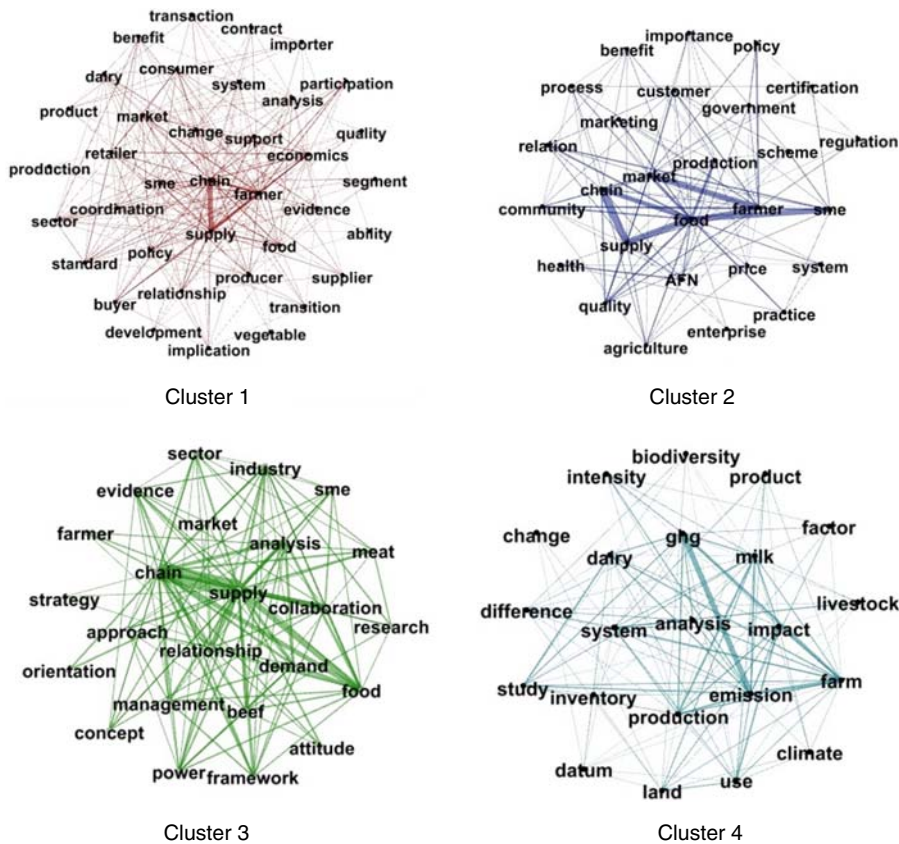


Figure 4.
Gephi visualization
results of text mining
using StanfordNLP
within each cluster
(2002–2014)

Clusters 1 and 2 have received much attention since they include 75 and 60 papers, respectively, among the 228 articles. Clusters 3 and 4 have received considerable attention with 51 and 33 papers, respectively. Being a smaller sub-field, only nine articles are included in cluster 5. The top five articles in each cluster, based on at least one of the four measures of centrality, are given in Table VIII. Recent papers have appeared in Clusters 2–5, whereas the works of {431} Ilbery *et al.* (2006a), {419} Higgins *et al.* (2008), {417} Risku Norja *et al.* (2008) and {381} Freidberg *et al.* (2011) of 2002–2014 appeared in the first cluster, signifying their influence in the complete network. This is due to the fact that BC is able to cluster very recent articles but it clusters few very old research papers (Boyack and Klavans, 2010). Among the recent articles, the works of {227} Forssell *et al.* (2015), {187} Giampietri *et al.* (2016a) and {169} Kulak *et al.* (2016) have a high degree closeness and eigenvector centrality. The betweenness centrality is found to be high for {291} Tsolakis *et al.* (2014), {275} Freidberg (2014) and {379} Fischer *et al.* (2009), which serves as a bridge in the whole network connecting different clusters. Similarly, the eigenvector centrality of the articles in the first cluster is higher, followed by {169} Kulak *et al.* (2016) and {225} Kulak *et al.* (2015) of Cluster 3. Cluster 5 has the least eigenvector centrality in which the lead articles have an eigenvector centrality of 0.0752, indicating that the research issue of these clusters has less association with other clusters in the network. To identify the research theme in each cluster, we further conducted a TM technique and established a co-word network as shown in Figure 6.

Table VII.
Top 10 co-occurred
words in the clusters
based on centralities
during 2002–2014

Cluster No.	Co-occurred words	Degree	Closeness	Centrality Betweenness	Eigenvector
1	Farmer	48	0.9000	176.77	0.9731
	Chain	48	0.9000	140.09	1.0000
	Supply	46	0.8710	128.30	0.9689
	Food	34	0.7297	60.02	0.7640
	Economics	33	0.7200	49.16	0.7729
	SME	38	0.7714	72.11	0.8471
	Change	27	0.6667	22.68	0.6703
	Market	33	0.7200	56.62	0.7522
	Contract	13	0.5684	7.04	0.3335
	System	24	0.6429	17.39	0.6048
2	Food	53	0.8955	457.70	1.000
	Farmer	35	0.7059	130.77	0.775
	Market	27	0.6452	69.19	0.627
	Production	27	0.6452	53.58	0.617
	Supply	29	0.6593	54.50	0.694
	Chain	32	0.6818	85.16	0.741
	AFN	62	0.8284	285.34	0.812
	SME	44	0.7357	89.64	0.736
	Marketing	21	0.5769	27.43	0.555
	Quality	16	0.5714	10.55	0.448
3	Supply	28	1.0000	76.25	1.000
	Chain	27	0.9655	67.89	0.976
	Relationship	18	0.7368	16.73	0.748
	Food	16	0.7000	10.26	0.700
	Collaboration	11	0.6222	1.81	0.533
	Industry	18	0.7368	14.02	0.760
	Approach	15	0.6829	8.77	0.653
	Sector	16	0.7000	10.82	0.682
	Management	12	0.6364	3.09	0.570
	Framework	11	0.6222	2.94	0.516
4	Emission	23	0.8966	34.35	1.0000
	Farm	20	0.8125	19.71	0.9204
	GHG	18	0.7647	14.49	0.8524
	Impact	20	0.8125	21.86	0.9054
	Production	16	0.7222	8.52	0.7982
	System	16	0.7222	6.94	0.8023
	Milk	22	0.8667	28.85	0.9677
	Land	14	0.6842	12.94	0.6446
	Use	18	0.7647	23.15	0.8139
	Livestock	8	0.5909	0.34	0.4646

The top 10 co-occurred words in each cluster are identified, and results are presented in Table IX. The degree measure shows that the production and distribution in AFN, relationship, safety and standards in FSC, greenhouse gas (GHG) emission impact of farm food system, traceability and product quality in FSC and asymmetric price transmission in the FSC are the research concepts of Clusters 1, 2, 3, 4 and 5, respectively. It is observed from Table IX that Supply, Chain, Farmer, Food and Market are the bridge keywords as they are found in more than one cluster. Also, Clusters 1 and 2 have more bridge keywords with high betweenness centrality indicating their close association. The keyword “SME” appeared only in the first and fourth cluster, based on co-occurrence, as shown in Table IX. However, from Figure 6, it is observed that “SME” is associated with Clusters 2 and 3, indicating the association of this keyword with the corresponding clusters. It is also observed that the

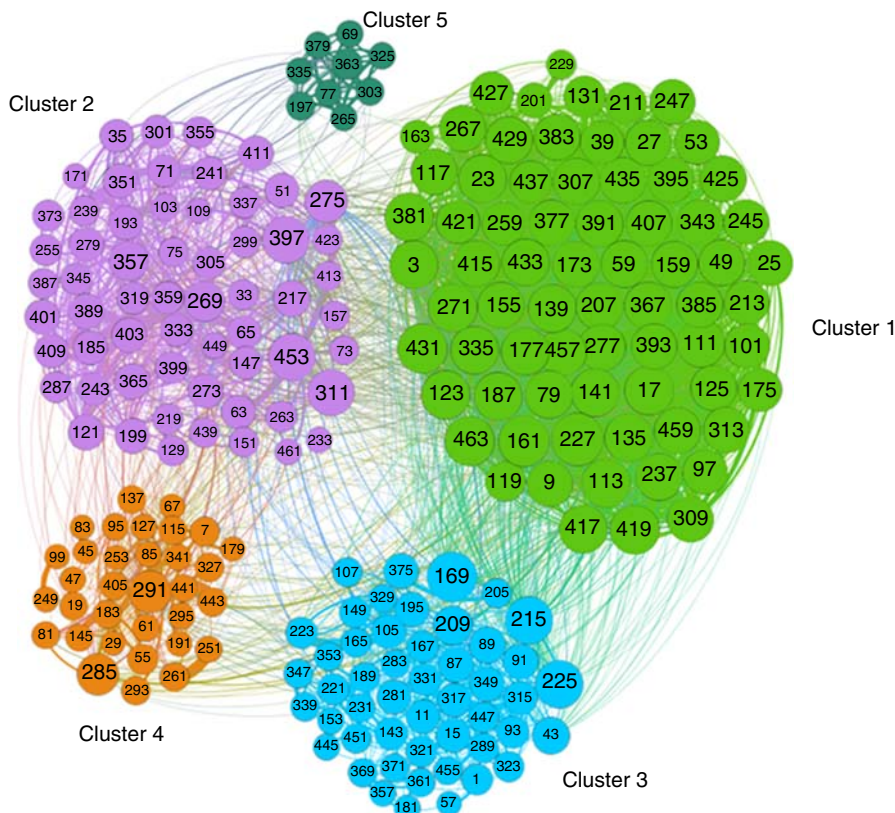


Figure 5.
Clustering of the
bibliographic coupled
articles based on the
modularity index
using Gephi
(2002–2018)

eigenvector and degree centralities of the word SME in Clusters 1 and 2 are high, albeit its low co-occurrence frequency compared to the previous period.

The historical relations of the research themes emerged during the periods 2002–2014 and 2002–2018, as shown in Figure 7. The broader thematic context is obtained for the period 2002–2014, whereas the inclusion of recent articles (i.e. 2015–2018) resulted in identifying the emerging research directions out of these broader issues. Through this approach, we are able to identify the dynamic evolution of the research domain. The first cluster “Production and distribution in AFN” has emerged from the Cluster 2 of period 2002–14, as shown in Figure 7. In recent years, AFN research is focussed on online spaces for reconnection (Bos and Owen, 2016), consumer behaviour towards buying in short food supply chains (SFSC) (Giampietri *et al.*, 2016a), trust and relationship in AFN (Roy *et al.*, 2017; Pinna, 2017), farmer’s market (Schupp, 2017), economic sharing in AFN (Miralles *et al.*, 2017) and so on.

The second cluster “Relationship, safety, and standards in the food supply chain” has emerged from Cluster 3 of 2002–2014, which focussed on improving the performance of the supply chain through the relationship among members. Recent articles in this sub-theme are focussed on retailers’ adoption of safety standards and its impact on organizations sourcing methodology. It is observed that the publications on the contractual relationship among supply chain members from the first cluster (market economics of contract farming in the FSC) during 2002–2014 are clustered with this subgroup (e.g. Reardon *et al.*, 2014; Poulton *et al.*, 2010).

Table VIII.
The values and ranks
[in bracket] of top 5
articles based on at
least one of the four
measures of centrality
during the period
2002–2018

Cluster	Article[ID]	Degree	Closeness	Centrality	
				Betweenness	Eigenvector
1	Forssell <i>et al.</i> (2015) [227]	91 [1]	0.6045 [1]	581.46 [1]	1 [1]
	Ilbery <i>et al.</i> (2006a) [431]	84 [2]	0.5691	262.77 [4]	0.9767 [2]
	Giampietri <i>et al.</i> (2016a) [187]	80 [4]	0.5815 [2]	250.69 [5]	0.9725 [3]
	Oglethorpe <i>et al.</i> (2013) [335]	77	0.5706 [5]	67.32	0.9661 [4]
	Higgins <i>et al.</i> (2008) [419]	83 [3]	0.5752 [3]	210.25	0.9643 [5]
	Freidberg <i>et al.</i> (2011) [381]	79 [5]	0.5633	293.91 [3]	0.9642
	Bos <i>et al.</i> (2016) [139]	77	0.5706 [4]	181.53	0.9612
	Risku Norja <i>et al.</i> (2008) [417]	79 [5]	0.5519	352.28 [2]	0.9289
	Fischer <i>et al.</i> (2009) [397]	79 [1]	0.5799 [1]	1003.00 [2]	0.7412 [1]
	Gómez <i>et al.</i> (2013) [311]	72 [2]	0.5768 [2]	749.81 [3]	0.5328 [2]
2	Freidberg (2014) [275]	63 [3]	0.5558 [3]	1362.04 [1]	0.2562 [3]
	Trifković (2014) [269]	54 [4]	0.5417 [4]	334.19 [5]	0.2078 [4]
	Bloom (2015) [217]	41 [5]	0.5095	108.75	0.1984 [5]
	Brooks <i>et al.</i> (2017) [51]	35	0.5165 [5]	273.78	0.1624
	Touboul <i>et al.</i> (2014) [273]	33	0.5082	393.25 [4]	0.1444
	Kulak <i>et al.</i> (2016) [169]	84 [1]	0.5863 [1]	601.90 [2]	0.8829 [1]
	Kulak <i>et al.</i> (2015) [225]	78 [2]	0.5487 [3]	330.26 [5]	0.8608 [2]
	Torquati <i>et al.</i> (2015) [215]	76 [3]	0.5706 [2]	665.20 [1]	0.7971 [3]
	Theurl <i>et al.</i> (2017) [43]	45 [5]	0.5119 [5]	265.34	0.3396 [4]
	Keyes <i>et al.</i> (2015) [209]	50 [4]	0.5194 [4]	394.82 [3]	0.3308 [5]
3	Repar <i>et al.</i> (2017) [89]	37	0.4628	330.54 [4]	0.1768
	Bourlakis <i>et al.</i> (2014) [285]	52 [2]	0.5459 [2]	890.16 [2]	0.4631 [1]
	Tsolakis <i>et al.</i> (2014) [291]	59 [1]	0.5572 [1]	1563.23 [1]	0.2579 [2]
	Banterle <i>et al.</i> (2014) [293]	17 [5]	0.4458	129.73	0.1480 [3]
	Lefebvre <i>et al.</i> (2014) [261]	17 [5]	0.4602 [5]	109.76	0.0777 [4]
	Abd Rahman <i>et al.</i> (2016) [55]	21 [3]	0.4830 [3]	277.00 [3]	0.0633 [5]
	Manzini <i>et al.</i> (2013) [327]	18 [4]	0.4618 [4]	202.88 [5]	0.0520
	Giagnocavo <i>et al.</i> (2017) [19]	14	0.4476	249.50 [4]	0.0375
	Aysoy <i>et al.</i> (2015) [197]	14 [2]	0.4155 [1]	505.96 [1]	0.0752 [1]
	Bakucs <i>et al.</i> (2012) [363]	23 [1]	0.4015 [2]	190.21 [2]	0.0315 [2]
4	Bakucs <i>et al.</i> (2014) [303]	13 [3]	0.3566 [4]	48.38 [3]	0.0138 [3]
	Dai <i>et al.</i> (2017) [69]	10 [4]	0.3620 [3]	28.75 [4]	0.0074 [4]
	Pokrivcak <i>et al.</i> (2014) [265]	9 [5]	0.3292 [5]	13.78 [5]	0.0058 [5]
5					

Cluster 3 “GHG Emission impact of farm food system” included 45 articles and it received great attention in recent years. This subgroup has emerged from Cluster 4 “Land use and emission impact of livestock systems” of 2002–2014. Recent articles in this subgroup are related to the emission impact of different types of food products other than livestock such as seafood (Jennings *et al.*, 2016; Ziegler *et al.*, 2016), fruits and vegetables (Lee *et al.*, 2015; Theurl *et al.*, 2017), meat and poultry (Wiedemann *et al.*, 2015a; Cottle *et al.*, 2016) and so on.

Cluster 4 “Traceability and product quality in the food supply chain” has received much attention in recent years, as food scandals and scares all over the world have resulted in a reduced consumer’s confidence. Recently in Chennai, the capital of the Indian state of Tamil Nadu, nearly two tonnes of meat, which was meant for supply to hotels in the city, was seized by the food safety officials, as it was suspected to be that of dog (*The Times of India*, 2018). The food traceability system is required to reduce the numerous cases of food safety incidents and fraudulence. Designing and implementing effective food traceability systems is a complex process, which requires an understanding of not only technical issues, but also of business, organizational, economic and social aspects (Giagnocavo *et al.*, 2017). Innovations in ICT that support research on traceability and food safety need attention. Cluster 5 “Asymmetric price transmission in food supply chain” has received considerable attention in the agro-food sectors (Von Cramon Taubadel, 2017). It is observed that among

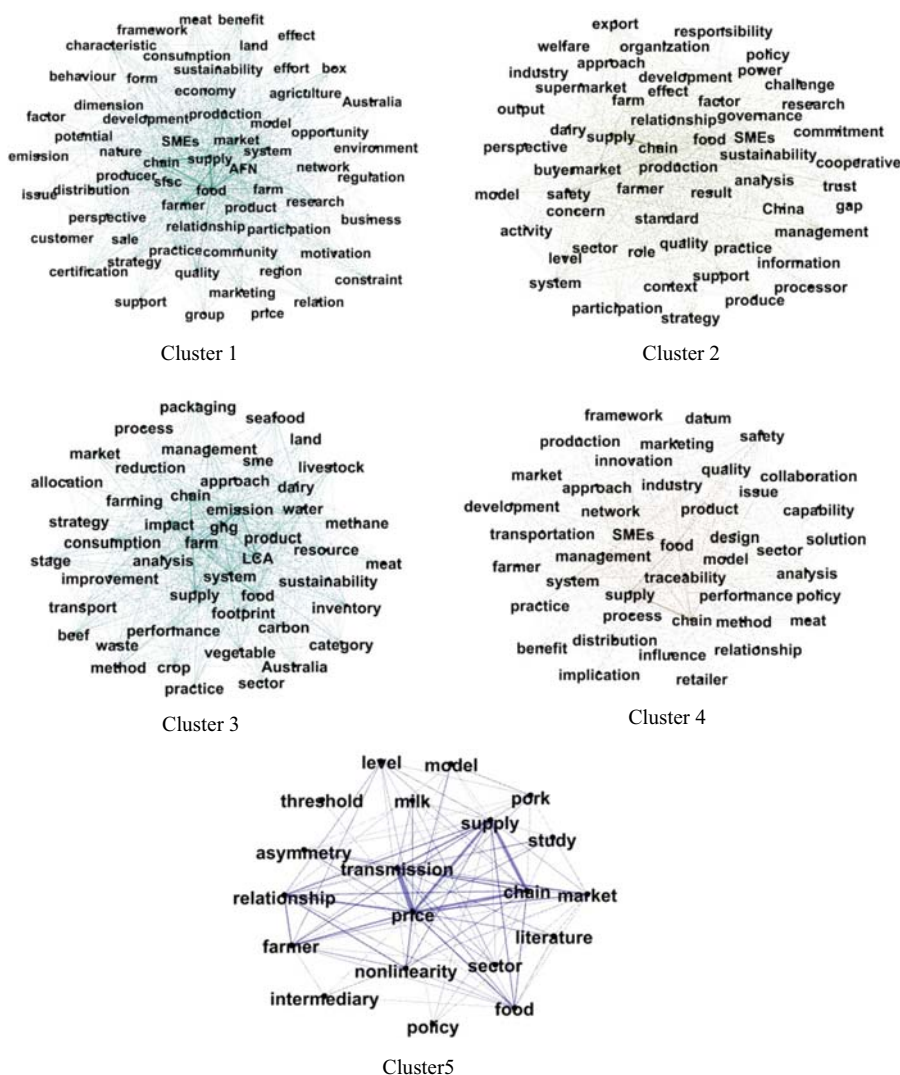


Figure 6.
Gephi visualization
results of text mining
using StanfordNLP
within each cluster
during the period
2002–2018

1,091 shortlisted articles, only 24 research papers focussed on price transmission in the FSC, whereas majority of the articles reported the effect of crude oil price and variation in food prices (Serra and Zilberman, 2013; Nazlioglu, 2011), impacts of animal disease crisis on price transmission (e.g., Park *et al.*, 2008) and so on. Since we included only those articles that had at least seven couplings with minimum of two other articles, the number of articles in this cluster is reduced to 9 most interconnected papers. Aramyan and Kuiper (2009) presented a conceptual overview of price transmission within agri-FSCs.

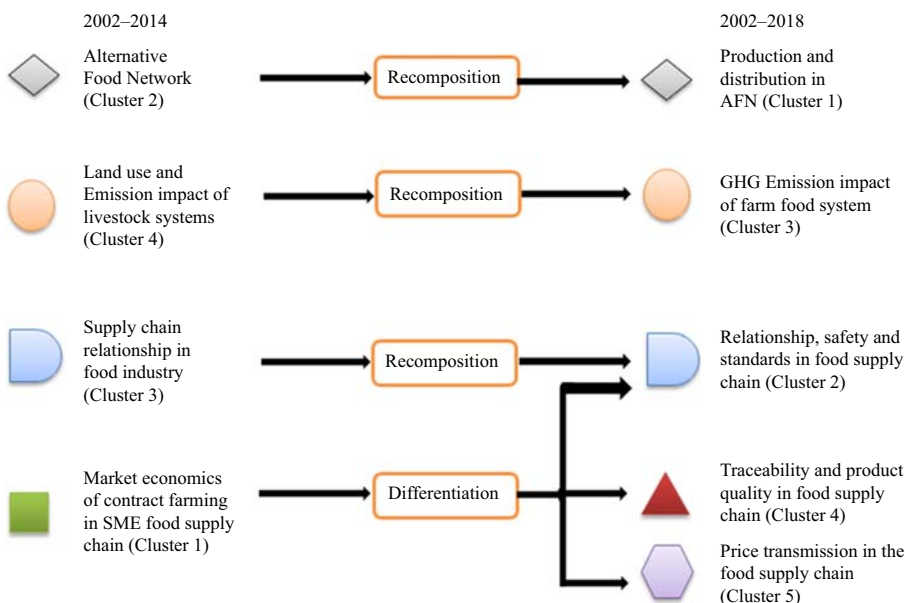
3.3 Thematic analysis

The research articles in the entire time period (i.e. from 2002 to 2018) are further analysed according to the research focus, the food sector studied and solution

Cluster No.	Label	Degree	Closeness	Centrality	
				Betweenness	Eigenvector
1	Food	120	0.9852	628.46	1.0000
	Farmer	106	0.9107	376.99	0.933
	Market	96	0.8452	257.25	0.8587
	Supply	91	0.8098	221.48	0.8684
	System	89	0.8598	230.37	0.8425
	Chain	89	0.8158	213.13	0.8587
	AFN	78	0.6178	39.21	0.4734
	SFSC	78	0.7157	116.46	0.6688
	Production	76	0.7257	151.86	0.7092
	SMEs	72	0.7211	148.63	0.7025
2	Supply	94	0.9799	322.73	1.0000
	Chain	93	0.9600	286.55	0.9980
	Farmer	82	0.8736	199.37	0.9128
	Food	77	0.8289	145.30	0.8452
	Relationship	72	0.7883	157.48	0.8176
	Safety	58	0.7157	71.59	0.6657
	Standard	58	0.7178	66.76	0.7148
	Sustainability	53	0.6678	62.64	0.6245
	Market	60	0.7158	104.20	0.7125
	Farm	56	0.7025	56.96	0.678
3	Emission	86	0.8750	158.53	0.9112
	Impact	85	0.8596	133.11	0.9120
	Farm	84	0.8750	157.25	0.9124
	LCA	83	0.9158	193.73	0.9555
	GHG	80	0.8305	127.29	0.8636
	System	79	0.8305	123.11	0.8714
	Product	74	0.7903	105.20	0.8069
	Food	72	0.7313	68.75	0.7258
	Footprint	70	0.7840	90.13	0.8043
	Chain	47	0.6577	26.51	0.5950
4	Supply	68	0.9048	217.55	0.9844
	Chain	68	0.9048	217.55	0.9844
	Food	71	0.9383	284.59	1.0000
	SME	51	0.7525	108.18	0.7917
	Product	49	0.7379	103.15	0.7486
	Traceability	40	0.6786	38.04	0.6882
	System	44	0.7037	62.06	0.7166
	Quality	49	0.7379	105.07	0.7669
	Model	39	0.6726	60.25	0.6142
	Analysis	37	0.6609	60.71	0.5991
5	Price	71	0.9141	221.45	0.9844
	Transmission	71	0.9224	227.65	0.9844
	Supply	70	0.9138	274.58	1.0000
	Chain	61	0.8524	208.58	0.7916
	Market	58	0.7937	203.41	0.7486
	Farmer	58	0.7857	118.94	0.6882
	Food	54	0.7097	112.10	0.7165
	Relationship	50	0.7137	105.07	0.7669
	Asymmetry	48	0.6895	71.42	0.6142
	Nonlinearity	47	0.6660	70.170	0.5990

Table IX.
Top 10 co-occurred
words in the clusters
based on centralities
during 2002–2018

methodologies used. The solution approaches of this time window are further grouped as quantitative models (includes analytical, simulation and network models), conceptual and qualitative models (describe and develop methods/frameworks for food SME supply chain), case study/empirical models (uses data collected from specific industry or from a



Supply chain issues in SME food sector

35

Figure 7.
Development pattern of SC issues in SME food sector

qualified source) and review works, as shown in Table X. From Table X, it is evident that case study/empirical modelling, followed by conceptual or qualitative techniques, are the most commonly used solution methodology to address supply chain issues in SME food sectors.

The different research approaches and various data analysis methods are shown in Figures 8 and 9, respectively. The most popular solution methodology used is the case study/empirical method (41.22 per cent), followed by conceptual/qualitative (36.4 per cent), literature review (15.3 per cent) and quantitative methods (7 per cent), respectively. Also, it is identified that a survey is the most commonly used empirical method. Data analysis methods, for instance, theoretical description, life cycle assessment (LCA), simulation, structural equation model, factor analysis, etc., have been extensively used. The most popular data analysis approach is found as the theoretical description, followed by a survey method. Our results agreed with the findings of Prakash (2018).

The distribution of research papers on the basis of food sectors is shown in Figure 10. Following the previous literature, we identified five major SME food sectors, which are classified as dairy, fruits and vegetables, meat and poultry, seafood and grains and oilseed. The dairy sector is the most addressed (36 per cent), followed by fruits and vegetables (27 per cent), meat and poultry (18 per cent), seafood (10 per cent) and grains and oilseed (8 per cent). It is also identified that the dairy sector has received significant attention in “GHG Emission impact of farm food system” and “Relationship, safety, and standards in food supply chain” themes. Fruits and vegetables have received significant attention in all the identified themes. Meat and poultry sectors have received much attention in traceability and product quality in the FSC and “Relationship, safety and standards in food supply chain” themes. Sea foods and grain and oil seeds sectors have received limited attention among the identified themes. It is observed that the dairy sector, followed by fruits and vegetables and meat and poultry, have received much attention in the academic literature, whereas seafood and grains and oilseed need further research.

Table X.
Solution methodology
and food sector
addressed in the
reviewed articles

Cluster	Case study/ empirical	Solution methodology		Quantitative	Dairy	Meat and poultry	Food sector		Grain and oil seeds
		Conceptual/ qualitative	Review				Fruits and vegetables	Seafood	
Production and distribution in AFN	45	25	6	3	1	2	5	0	1
Relationship, safety and standards in food supply chain	17	30	11	0	12	3	9	3	3
GHG emission impact of farm food system	29	8	8	5	20	5	6	6	3
Traceability and product quality in food supply chain	3	13	8	8	0	6	5	0	1
Asymmetric price transmission in the food supply chain	0	7	2	0	2	2	2	1	0
Total	94	83	35	16	35	18	27	10	8

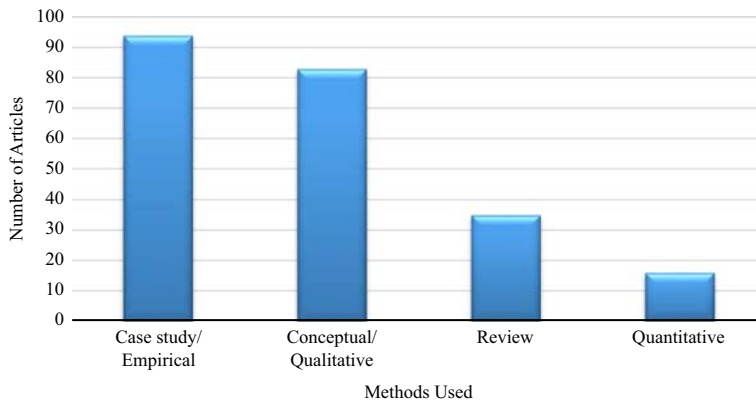


Figure 8.
Pattern of
research methods

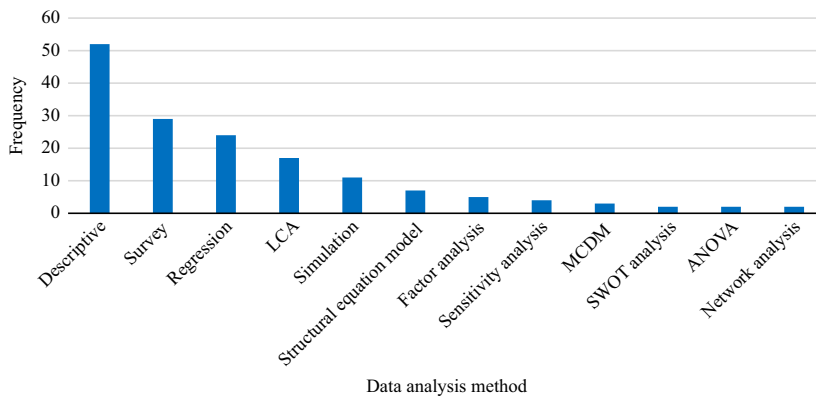


Figure 9.
Pattern of data
analysis method

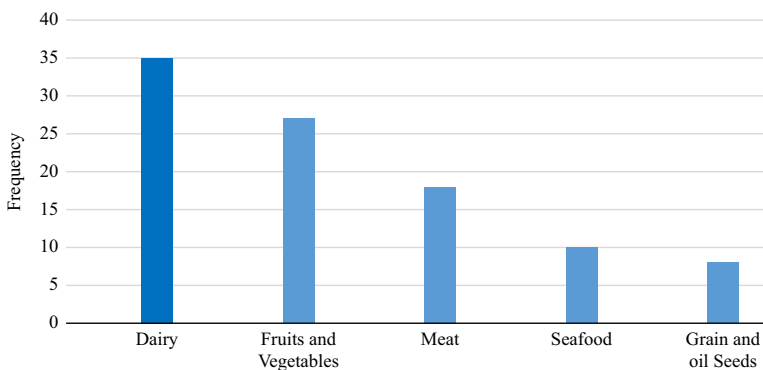


Figure 10.
Type of food products

3.3.1 Cluster 1: production and distribution in alternative food network (AFN). AFN may be defined as the networks of production, distribution and consumption of food that propose and practice models that can be considered as an alternative to the ones of the conventional/standardized food supply system (Bazzani and Canavari, 2013). AFNs are defined by the

characteristics like spatial relationship between farmers and consumers (Jarosz, 2008), SFSC, for example, farmers' markets (Wills, 2017; Giampietri, 2016a; Miralles *et al.*, 2017; Aggestam, 2017), community-supported agriculture (CSA) (Barta, 2017; Pascucci *et al.*, 2011; Syrovátková *et al.*, 2015) and box scheme (Miralles, 2017; Freidberg, 2011), the existence of "alternative" food purchasing models and retail venues, usually based on human relationships and proximity between consumers, producers and/or retailers; small farm size and scale or environmentally sustainable farming methods, and a commitment to the social, economic and environmental dimensions of sustainable food production, distribution and consumption.

SFSCs shift food production out of "industrial modes" with long and complex structures to a shorter system in which consumers are closer to the origin of their food. SFSCs enable SMEs to capture a higher proportion of value added, as well as making direct connections with final consumers. Farmers' markets, CSA and box schemes are some forms of SFSC through which local food products are made available to consumers through a direct exchange. Farmers' markets deliver local food products, as well as knowledge about the food system and agricultural awareness to the consumers (Garner *et al.*, 2018). However, it is difficult for the farmers in the SME sector to deliver a steady, diverse and abundant quantity of local food products in such a way that it would eclipse the need for a globalized food system (Pearson *et al.*, 2011). CSA, in a sense, attempts to re-structure the food production and consumption by providing more importance to members of a community than conventional economic roles such as food producers and their consumers (Hinrichs, 2000). Box schemes represent the more common form of direct selling in the UK. It involves the exchange of pre-packed food products between suppliers and consumers (Zorbas *et al.*, 2018). Fruit and vegetable box schemes are attaining innovative, profitable niche markets since the demand for local food from the SME sector is dramatically increasing. Box schemes may denote one of the food system solutions in which food is re-localized from the market to farm and relations are personalized through transparency and broad contextual information directly from the food SMEs (Torjusen *et al.*, 2008).

It is observed that "CSA" have received much attention, whereas the research on farmer's market and "box scheme" is very limited. Conceptual or qualitative method is the most prominent solution methodology, followed by empirical method, whereas quantitative methods need further attention. From Table X, it is clear that most of the AFN research is focussed on the fruits and vegetable sector, followed by the meat and poultry sector.

3.3.2 Cluster 2: relationship, safety and standards in food supply chain. The relationship management includes the process of developing profound and long-standing relationships with consumers and suppliers through constructing links between the supply chain members (Fredriksson and Liljestrand, 2015). Although a number of research works are conducted on supply chain relationship in the food sector, the majority of them are focussing on the farmer–retailer or farmer–processor relationships. The relationship issues are further classified as contract-based relationship, a relationship grounded on trust and communication.

It is evident that contractual relationship among the supply chain members will decrease the transaction costs and market imperfections (Neilson, 2007), and intensify access to marketing (Blandon, 2010). Trust is considered as the soul of supply chain management (Fu *et al.*, 2017a), and it is the inter-personal belief acquired from the experience by enabling a safe and secure food supply (Fischer, 2013). According to Fu *et al.* (2017a), a trust-based relationship among the supply chain members can stimulate the information sharing and corporation. A successful SC relationship is defined as a collaborative relationship, characterized by a high level of trust and communication (Jonsson and Gustavsson, 2008). The quality of a food sector depends not only on its own quality but also on the quality of its suppliers, and hence the information sharing between the firms and their suppliers have an important role in the FSC (Huo *et al.*, 2018). Apart from this, the unclear data about the quality of the product will lead to a crisis, and

high-information quality can stimulate SC performance through improving the relationship (Jonsson and Gustavsson, 2008).

The safety issues are further divided into food safety standards and food safety management system (FSMS). For the growth of the domestic food retail sector and high-value exports, food safety standard is an essential factor (Unnevehr, 2015; Shukla *et al.*, 2018). Jaccsens *et al.* (2015) classified food safety and quality standards as European Union legislation, Codex Alimentarius standards, private standards and national legislation. To report food safety, every organization in the global food SC desires to implement a FSMS, and each FSMS is organizational specific since it is a result of the implementation of various legal requirements and quality assurance into a company's unique environment and production (Kirezieva *et al.*, 2015).

Luo *et al.* (2018) identified driving forces and barriers to the supply chain relationship in the SME food sector. The driving forces include food safety scandals, changing consumer demand and SC risks. Also, barriers consist of coordination difficulties due to the number of connecting entities, incomplete information sharing, malfunctions originating from diverse strategic planning practices, lack of trust among stakeholders, different entrepreneurial mentalities and failure to understand opportunities in the SME food sectors. Our result reveals that a case study or empirical methodology is the most commonly used solution methodology, followed by qualitative methods, whereas quantitative models need further attention. Dairy and fruits and vegetables sectors have received much attention, whereas other food sectors are not much addressed in this cluster.

3.3.3 Cluster 3: GHG emission impact of the farm food system. The agricultural production has developed significantly over the recent decades, resulting in the increased use of fertilisers and mechanization that lead to the environmental consequences (Tasca *et al.*, 2017). To assess the environmental impacts of food systems, various metrics have been established that include the quantification of GHG emissions and water usage of food items, assessment of water footprint and food production systems through LCA (Green *et al.*, 2018). Here, we classified the food emission according to GHG emission and water footprints. GHGs can be explained with different types of emissions including carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O) (Vida *et al.*, 2017; Christie *et al.*, 2011). The food products in the supply chain produce GHG emission from the farm-to-fork through manufacturing, distribution and even retailing (Dorward, 2012). Water pollution and consumption at a global scale are recognized as the key resource use issues. Agricultural production is the foremost water user, and the major water footprint approaches applied to agricultural commodities are the water footprint network method and LCA method (Zonderland-Thomassen *et al.*, 2012). The water footprint is characterized by green, blue and grey water along the whole life cycle of a product (Payen *et al.*, 2018). The green water footprint is the volume of water evaporated from the root zone of the soil, and it is particularly relevant for forestry, horticultural and agricultural products. The blue water footprint is the volume of freshwater that evaporated from surface or groundwater resources to produce goods and services. Domestic water use, irrigated agriculture and industry can each have a blue water footprint. Grey water footprint is the amount of freshwater required to dilute a load of pollutants to meet the specific water quality standards.

Our result indicates that 90 per cent of the research is on GHG emission studies and the water footprint issues need further attention. The conceptual methodologies have been widely used, whereas the empirical models and quantitative techniques investigating emission from the food products and its impact are limited and need further attention. Besides, our result indicates that the dairy sector is the most addressed food sector in this research theme, whereas research on fruits and vegetables, meat and poultry and seafood sector is limited.

3.3.4 Cluster 4: traceability and product quality in the food supply chain. Olsen and Borit (2013) defined traceability as the ability to access information related to things that are to be considered throughout their whole life cycle using recorded identification. The term “Traceability” includes both “tracing” and “tracking” (Dandage *et al.*, 2017). Tracing is a reverse process in which origin is identified in the supply chain, and tracking is a forward process where the end users are identified by the location in the supply chain.

The articles in this cluster are divided into three main sub-themes such as quality and safety, sustainability and cost-related issue. Traceability is recognized as a mechanism for quality and safety (Manzini *et al.*, 2013), improving sustainability (Giagnocavo *et al.*, 2017; Mattevi and Jones, 2016a) and reducing the overall cost (Dandage *et al.*, 2017) of the food product since it reduces the recall's probability. Dabbene *et al.* (2014) identified many issues related to traceability of bulk products, fraud prevention, quality and identity preservation concerns and food crisis management. For the past few years, many developed and developing countries, for instance, Austria, Germany, the UK, etc., have made it mandatory to implement food traceability systems (Riviere and Buckley, 2012). However, it is surprising that India, the second largest producer of fruits and vegetables in the world (Saxena and Gandhi, 2014), does not have any required traceability system (Schroeder and Tonsor, 2012).

In recent years, an increasing effort in the implementation of traceability systems can be seen, mainly as a result of a growing concern with food quality and safety issues (Saltini *et al.*, 2013). An optimized and sustainable supply chain necessarily requires taking into account the traceability (Germani *et al.*, 2015). To overcome the food quality and safety issues, it is necessary to develop a trusted food traceability system, which can track and monitor the complete lifespan of food production (Lin *et al.*, 2018). Of late, McDonald restaurants presented the food traceability system to monitor the item sources from various providers of the nation, keeping in mind the end goal to give food security and quality (Dandage *et al.*, 2017). For this, a mobile application called TrackMyMacca's was created. Through this application, the global positioning system (GPS) provides the location of the restaurant, the tracking image on the McDonald's packaging is recognized by the software, and combining these data along with date and time, the application accesses McDonald's supply database to get the story of the products. Proper traceability in the food industry is required due to the precarious nature of the existing supply chain and to reduce the numerous cases of food safety incidents and fraudulence (Dandage *et al.*, 2017). Improving sustainability in the SME food sector usually leads to a higher cost. However, by creating transparency through the traceability, consumers have to be convinced that often higher prices are involved to improve sustainability (Khan *et al.*, 2018). A smart traceability system in the SME food sector can be built by the application of modern technologies such as cloud computing, GPS, Internet of Things (IoT), Big Data, artificial intelligence (AI), etc., into traditional agriculture. Lin *et al.* (2018) conducted a study on the influence of Blockchain and IoT technologies on the agri-food sector and found that these technologies help to build a self-organized, open, trusted and ecological smart agriculture system, which includes all parties in the system, even if they may not trust each other.

From the analysis, it is observed that studies on the impact of traceability on sustainability and cost of the product need further attention. Developing smart traceability system in the SME food sector is a ripe research area. From Table X, it is observed that empirical and quantitative models have received much attention. Our results indicate that meat and poultry and fruits and vegetables sectors are the most addressed food sector in this cluster.

3.3.5 Cluster 5: asymmetric price transmission system in the supply chain. This cluster focussed on the degree to which price changes are transferred along the FSC. Price transmission,

in short, is the process in which upstream prices affect downstream prices. The welfare of farmers and food consumers is affected by horizontal and vertical price transmission in the FSC (Pokrivcak *et al.*, 2014). Price transmission from producer prices to consumer prices refers to vertical price transmission, whereas horizontal price transmission means the interconnection among different markets at the same position in a supply chain. The main causes of asymmetric price transmission, as narrated in the literature, are existence of adjustment and menu costs that occur due to the re-pricing and the acquisition of an advanced pricing strategy (Meyer and Cramon-Taubadel, 2004), market power (Dai *et al.*, 2017; Aysoy *et al.*, 2015; Bakucs *et al.*, 2014) and miscellaneous causes such as asymmetric information, inventory management and political intervention (Meyer and Cramon-Taubadel, 2004), etc. Adjustment cost occurs when there is a change in a new pricing strategy or re-pricing, and APT arise if these costs are asymmetric with respect to increases or decreases in prices (Simioni *et al.*, 2013). APT is mainly related to market distortion like market power of retailers or processors (Pokrivcak *et al.*, 2014). Generally, food industries are oligopolistic in nature (Bakucs *et al.*, 2014), and the price transmission is influenced more by the market power in downstream sectors by decreasing the purchasing prices in the upstream sectors.

Domestic producer prices are associated with the law of one price to world prices, whereas the consumer prices depend on both producer prices as well as the efficiency of price transmission (Liu *et al.*, 2018). In agricultural and food markets, a hike in the producer prices is transferred more rapidly to consumer prices than a producer price decline, which is passed through the FSC to consumer prices more slowly and incompletely. Empirical studies illustrate that APT is the rule rather than the exception (Rezitis and Pachis, 2018). Our results found that empirical modelling is widely used in this cluster.

3.4 Research implications

The food industry represents one of the largest manufacturing sectors in the global economy. FSC has been coined to depict the activities or operations from production, distribution, and consumption so as to keep the safety and quality of various food products under efficient and effective modes (Blandon *et al.*, 2009b). A major portion of the food processing and distribution industries participating in the FSC comprises of SMEs. Implementing SCM practices to improve the operations in food SMEs is not as easy as in bigger counterparts, as there may be lots of complexities and difficulties in analysing the operational objectives (Nabhani and Shokri, 2009). Indeed, both academics and practitioners pay increasing attention towards the supply chain issues in food SMEs. Despite this growing attention, the extant literature has yet to provide any evidence of a systematic review of supply chain issues in SME food sector. To address this research gap, we present an SLR of supply chain issues in SME food sector on the basis of BC and co-word analysis. The findings of this research exhibit the following implications for research, practice, and society. The policy makers could use the findings for developing strategies for the betterment of SMEs in the FSC, which will provide huge employment opportunities and ensure regional balance by taking industrialization to rural and backward areas.

Our review contributes to the body of FSC literature by reporting the major supply chain issues/themes, the current research trends and the different modelling approaches used in the SME food industry. We have identified five major themes, of which two themes, namely, GHG emission impact of the farm food system and traceability and product quality in the FSC, have received increasing attention in recent years. Our findings support the recent study (Mattevi and Jones, 2016a) that suggests managers should see “being traceable” as a strategic objective as it is “being green”, and it further reports that traceability could follow the sustainability development path, due to consumer pressure on traceability. Food traceability is one of the main factors in the evaluation of quality assurance systems in the

SME sector. A smart traceability system in the SME food sector can be built by the application of modern technologies such as IoT, cloud computing, GPS, Big Data, block chains, AI, etc., as the findings of Mattevi and Jones (2016b) confirmed that SMEs of the UK do not consider technology as a driver to implement traceability, albeit product safety, quality and regulations are observed as the main drivers. Several countries have set up the specific regulations or policies for food traceability of a broad range of food and animal products of both domestic and imported origin. Although traceability of livestock and meat products has improved through the harmonization of animal identification systems in many countries, systems allowing for the same level of traceability for other food commodities are still far from being fully implemented (Charlebois, 2015).

The FSC contributes 20–30 per cent of the GHG emissions in terms of the global environmental impact. This is mainly as a result of a large amount of resources required for the food production and formation of the food waste in the whole life cycle (Noya *et al.*, 2018). However, the consumers' awareness of high-quality food products produced in an environmentally friendly way has increased considerably in the last decade (Notarnicola *et al.*, 2012). In order to comply with consumers' expectations and also improve the competitiveness, companies and retailers have developed new strategic approaches for sustainable food production and consumption across the whole supply chain. LCA methodology has been extensively used as a tool to assess the environmental profile of the food products, as shown in Figure 9. The dairy sector has received increasing attention in the environmental impact analysis (see Table X). The strong environmental impact of milk production has driven researchers to investigate mitigation measures, and several countries have tried to respond with initiatives of varying levels of complexities at different stages of the supply chain (Torquati *et al.*, 2015). The concept of local food and food miles is gaining traction in recent years as a way of eating fresh and high-quality foods and reducing the environmental impact. When carbon footprints are exclusively observed, other important issues like water usage are neglected. Sustainable practices for water footprint mitigation in FSC need further attention.

4. Conclusion

This paper presents an SLR and a comprehensive analysis of 1,091 articles in supply chain issues in SME food sector from 2002 to 2018. To understand the dynamic evolution, we adopted a comparative approach to the analysis of the data, which we divided into two roughly equal parts. We used BC combined with co-word analysis to identify the historical relations of the research themes that emerged during the periods 2002–2014 and 2002–2018. The broader thematic context is obtained for the period 2002–2014, whereas the inclusion of recent articles (i.e. 2015–2018) resulted in identifying the emerging research directions out of these broader issues. From the results of bibliographic network and co-word analysis on the data, we detected five major themes as production and distribution in AFN, relationship, safety and standards in FSC, GHG emission impact of farm food system, traceability and product quality in FSC and asymmetric price transmission in the FSC. It is identified that GHG emission and traceability themes have received increasing attention in recent years. It is observed that “SME” is associated with all the four clusters except the asymmetric price transmission, indicating the association of this keyword with the corresponding clusters. The research articles are further analysed with reference to the research focus, modelling approaches used and food sectors addressed. The major food sector addressed in the SME FSC is the dairy sector, followed by fruits and vegetables, meat and poultry, seafood sector and grain and oil seeds, respectively. The most commonly used modelling techniques are empirical methodology/case study, followed by conceptual or qualitative method, and review; however, quantitative method needs further attention.

The proposed review has a few limitations. In our study, only the title and abstract of all the research articles in each cluster are text mined. Our results and findings may advance if the frequent words or phrases used in the entire research article are text mined. As a future extension and to avoid the drawbacks, the entire article can be included for the analysis. Unpublished works, conference proceedings and doctoral thesis were not included in the review. Also, we excluded articles before 2002, even though the number of articles published before this time period is very limited.

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Appendix 1. Network properties calculation

Degree centrality

One of the simplest local measures of centrality is the degree centrality. According to this measure of centrality, the most prominent nodes are the ones that have the most ties to other nodes in the network. The number of neighbours is defined as nodal degree or degree of connection. The degree

centrality is represented by the number of direct linkage between node i and other nodes, and it is given as follows:

$$\text{Degree centrality } C_i = \sum_j m_{ij},$$

where $m_{ij} = 1$ if node i and node j are linked in a network.

Betweenness centrality

The concept of betweenness is a measure of how often a node is located on the shortest path between other nodes in the network. Those nodes positioned on the shortest path between other nodes play the roles of mediator that influences any two nodes without direct contact. Nodes with higher betweenness centrality are those positioned at the core of the network:

$$\text{Betweenness centrality}(B_i) = \sum_{j,k \neq i} \frac{g_{jik}}{g_{jk}},$$

where g_{jk} is the shortest path between node j and node k ; g_{jik} the shortest path between node j and actor k that contain actor i .

Closeness centrality

The closeness centrality of a node is defined by the inverse of the average length of the shortest paths to/from all the other nodes in the network. Higher closeness centrality indicates higher influence on other nodes:

$$\text{Closeness centrality}(CC_i) = \sum_{j=1}^N \frac{1}{d_{ij}},$$

where d_{ij} is the shortest path between node j node i .

Eigenvector centrality measures

Consider an undirected network represented by a graph $G = (V, E)$, where V denotes the set of nodes $(1, 2, \dots, c, \dots, d, \dots, v)$ and E the set of edges. Let A be the co-occurrence matrix for this graph and its element $acd = 1$ (or edge weightage in case of weighted network) if nodes c and d are connected by an edge and $acd = 0$ if they are not.

Eigenvector centrality x can be defined in two equivalent ways via a matrix equation and a sum as $Ax = \lambda x$ and $\lambda x_c = \sum_{v=1}^V a_{cv} x_v$, respectively, where λ is the largest eigenvalue of A (Bonacich, 1987). This centrality of a node is proportional to the sum of centralities of all other nodes to which it is connected, that is a node that has a high eigenvector score is one that is adjacent to nodes that have high scores.

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